

# $p$ -adic Variational Quantum Circuits Have No Barren Plateaus

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# Agenda

- 1 The obstruction
- 2 Quantum computing over  $\mathbb{Q}_p$
- 3 Asking the question correctly
- 4 Main result
- 5 Proof sketch
- 6 Discussion

# Barren plateaus in variational quantum circuits

A variational circuit prepares  $U(\theta)\psi_0$  and tunes  $\theta$  to minimise

$$\mathcal{L}(\theta) = \langle U(\theta)\psi_0, M U(\theta)\psi_0 \rangle.$$

McClean et al. (2018)

If the circuit ensemble forms a unitary 2-design on  $U(2^n)$ , then

$$\mathbb{E}[\partial_\theta \mathcal{L}] = 0, \quad \text{Var}[\partial_\theta \mathcal{L}] = O(2^{-n}).$$

- The landscape is flat almost everywhere.
- Resolving a descent direction costs a number of shots exponential in the qubit count.

# Why it matters for quantum machine learning

- Barren plateaus are the main argument against **scalable** variational quantum machine learning.
- Later work traced them to the cost function (Cerezo et al. 2021) and to ansatz expressibility (Holmes et al. 2022).
- The known mitigations act on the **circuit**:
  - initialisation and warm starts,
  - local cost functions,
  - shallow or structured ansätze.

Every one of them keeps the scalars:  $\mathbb{C}$ .

# What does the theorem actually use?

Read carefully, the barren-plateau theorem rests on three ingredients:

- ① a **compact** ensemble, so a Haar *probability* measure exists;
- ② second-moment data supplied by unitary ***t*-designs**;
- ③ “small” measured in the **Archimedean** absolute value.

None of the three is a fact about quantum mechanics. All three are facts about  $\mathbb{C}$ .

*What happens if we change the field?*

# Changing the scalars

Following Aniello, Mancini and Parisi, rebuild quantum kinematics over the unramified quadratic extension  $K = \mathbb{Q}_p(\sqrt{\mu})$  of the  $p$ -adic field:

- states are vectors in  $K^N$ , with  $N = 2^n$ ;
- gates are unitaries for the Hermitian form  $\langle v, w \rangle = \sum_i \bar{v}_i w_i$ ;
- the geometry is **ultrametric**:  $|x + y|_p \leq \max\{|x|_p, |y|_p\}$ .

## Why anyone builds this

Ultrametric geometry is the intrinsic geometry of **hierarchical data**: taxonomies, phylogenies, file systems. Classical  $p$ -adic learning has used it for decades.

The quantum side already has a qubit, a tensor product, a universal gate set and a first quantum neural network.

# Fact 1: small means divisible

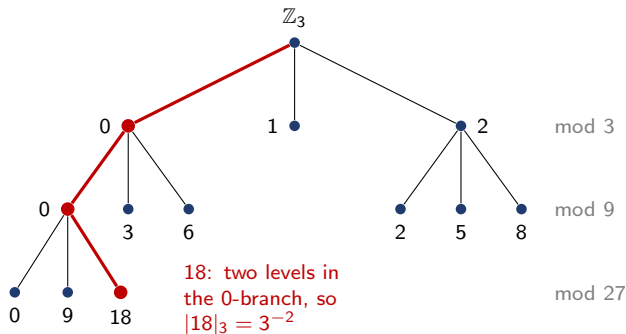
$|x|_p = p^{-v_p(x)}$ , where  $v_p(x)$  counts the factors of  $p$  in  $x$ .

At  $p = 3$ :

$$|18|_3 = \frac{1}{9} \quad (18 = 2 \cdot 3^2), \quad |2|_3 = 1.$$

Two numbers are close when they agree **modulo a high power of  $p$** : they sit deep in the same branch of the tree.

Distances are read off a hierarchy: this is why the geometry fits hierarchical data.



# Facts 2 and 3: a discrete scale, and sums that do not grow

## 2. The scale is discrete

$|\cdot|_p$  takes values in  $p^{\mathbb{Z}}$ . A gradient is a unit, or divisible by  $p$ , or by  $p^2$ , and nothing in between. There is no continuum of “slightly small” quantities.

## 3. Sums do not grow; cancellation is a congruence

$|x + y|_p = \max\{|x|_p, |y|_p\}$  whenever the two sizes differ. A sum is as large as its largest term *unless* the leading digits cancel, and that is a condition modulo  $p$ : finite and checkable.

Fact 3 is why the whole second-moment analysis will collapse onto **counting in finite groups**.

# One thing that does not carry over: sampling

- $K$  is not ordered, so  $\langle \psi, \psi \rangle \geq 0$  means nothing: states form an affine, not a convex, set.
- Generalised probabilities lie in  $\mathbb{Q}_p$ , not in  $[0, 1]$ . They have no frequentist reading and **cannot be estimated by repeated measurement**.
- Readout is algebraic: the algorithm returns  $p$ -adic numbers, read digit by digit.

## So what is a plateau here?

Over  $\mathbb{C}$  a small gradient is bad because it takes many shots to resolve. Over  $K$  there are no shots: a gradient of high valuation is bad because the *leading digits of the parameter update vanish*.

# Surprise 1: the usual ensemble does not exist

## Proposition

For every  $N \geq 2$  the unitary group  $U(K^N)$  is unbounded, hence non-compact, and carries *no* Haar probability measure.

- Explicit witness at  $p = 3$ ,  $K = \mathbb{Q}_3(i)$ : the vector  $u = (\frac{1}{3}, c)$  with  $\text{Nm}(c) = \frac{8}{9}$  has  $\langle u, u \rangle = 1$  but  $|\frac{1}{3}|_3 = 3$ . Witt's theorem completes it to a unitary of norm 3; replacing  $\frac{1}{3}$  by  $3^{-m}$  gives norm  $3^m$ .
- The reason is structural: over a local field  $\sum \bar{z}_i z_i$  is **isotropic**. Over  $\mathbb{C}$  compactness came from positivity, and positivity is what the  $p$ -adic model gives up.

“Haar-random circuit” over the full group has no meaning.

# The fix: integral unitaries

## Definition

$G_N = U_N(\mathcal{O}) = \{U \in U(K^N) : \text{all entries of } U \text{ lie in the ring of integers } \mathcal{O}\}.$

- $G_N$  is compact and carries a unique Haar probability measure  $\nu$ .
- It costs nothing for quantum computing: it contains every circuit built from the known  $p$ -adic gate sets:
  - Pauli gates and CNOT (entries in  $\{0, \pm 1\}$ );
  - the  $SO(3)_p$  gates of Svampa et al. (roots of unity);
  - the Hadamard gate, whenever  $\sqrt{2} \in K$ .

$G_N$  is the smallest natural compact group containing all realisable circuits, so it is our ensemble.

## Surprise 2: the benchmark is $p^{-1}$ , not 1

The series  $\exp_p(X) = \sum_m X^m/m!$  converges only for  $\|X\| < p^{-1/(p-1)}$ . Because the value group is discrete, this is **equivalent** to  $\|X\| \leq p^{-1}$ , that is  $X \equiv 0 \pmod p$ .

Hence every convergent one-parameter ansatz satisfies

$$\exp_p(\theta X) \equiv \mathbf{1} \pmod p \quad \implies \quad |\partial_\theta \mathcal{L}|_p \leq p^{-1} \text{ always.}$$

- We therefore write the generator as  $p\theta A$  with  $A$  integral: the factor  $p$  is not a normalisation we chose, it *is* the convergence condition.
- “As large as possible” means  $|\partial_\theta \mathcal{L}|_p = p^{-1}$ , i.e. the normalised gradient  $\nabla \mathcal{L} = p^{-1} \partial_\theta \mathcal{L}$  is a  $p$ -adic unit.

# The question, stated precisely

$\text{Var} = O(2^{-n})$  has no  $p$ -adic meaning. The faithful translation asks how often the gradient is *as large as it can be*.

## Question

Let  $\nu$  be Haar measure on  $G_N$ ,  $N = 2^n$ . Fix an integral observable  $M = M^*$ , an integral generator  $A = -A^*$ , an admissible  $\psi_0$  and a block  $U_B \in G_N$ , and set

$$U(\theta) = U_B \exp_p(p\theta A) U_A, \quad \theta \in \mathbb{Z}_p.$$

If  $U_A \sim \nu$ , does  $\Pr[|\partial_\theta \mathcal{L}|_p = p^{-1}]$  stay bounded away from 0 as  $n \rightarrow \infty$ , and for which pairs  $(M, A)$ ?

**Heuristic.** A Haar-random element of  $\mathbb{Z}_p$  is a unit with probability  $1 - \frac{1}{p}$ . So one expects the answer  $\approx 1 - \frac{1}{p}$ , whatever  $n$ .

# No barren plateaus

Let  $\hat{M} = U_B^* M U_B$ , and let  $\Gamma$  be the commutator  $[\hat{M}, A]$  reduced modulo  $p$ .

## Definition ( $r$ -non-degenerate)

$\Gamma$  has an orthonormal eigenbasis over  $\mathbb{F}_{p^2}$  with eigenvalues in  $\mathbb{F}_p$  (*splitting*), and no eigenvalue has multiplicity above  $N - r$ .

## Theorem

If  $\Gamma$  is  $r$ -non-degenerate and  $U_A$  is drawn from  $\nu$ , then

$$\Pr\left[|\partial_\theta \mathcal{L}|_p = p^{-1}\right] \geq 1 - \frac{1}{p} - 10 p^{1-r}.$$

The gradient attains the largest absolute value any convergent  $p$ -adic ansatz permits, with probability bounded below **independently of the number of qubits**. The same bound holds if  $U_B$  is Haar-random too.

# Reading the theorem

- **No dependence on  $n$ .** Over  $\mathbb{C}$ , Chebyshev gives  $\Pr[|\partial_\theta \mathcal{L}| \geq \varepsilon] = O(2^{-n} \varepsilon^{-2})$ . Over  $K$  the gradient is *maximal* with probability close to  $1 - \frac{1}{p}$ , for every  $n$ .
- **$r$  grows like  $2^n$ .**  $\Gamma$  has  $N = 2^n$  eigenvalues in a  $p$ -element field; spread over the available values,  $r = \Theta(2^n)$ .
- **Doubly exponential correction.** The error term  $10p^{1-r}$  is of order  $p^{-2^n}$ , smallest exactly where the complex plateau is worst: many qubits, expressive circuits.
- **When it bites.** The bound is non-trivial once  $p^{r-1} > 10p/(p-1)$ , which in the regime  $r \sim 2^n$  holds from three qubits on.

# A witness you would actually write down

Take  $p \equiv 3 \pmod{4}$ , so  $K = \mathbb{Q}_p(i)$ . On  $n$  qubits:

$$M = X_1, \quad A = -\frac{i}{2}(\mathbf{1} - Z_1), \quad U_B = \mathbf{1}.$$

In words: *measure Pauli  $X$  on the first qubit, rotate the first qubit about  $Z$ .*

- $C = [M, A] = Y_1$ , so  $\Gamma^2 = \mathbf{1}$ : eigenvalues  $\pm 1 \in \mathbb{F}_p$ , each of multiplicity  $2^{n-1}$ , with orthonormal eigenbases.
- Hence  $\Gamma$  is  $r$ -non-degenerate with  $r = 2^{n-1}$ .

At  $n = 10$ ,  $p = 3$

$$\Pr[\text{gradient maximal}] \geq 1 - \frac{1}{3} - 3^{-500}.$$

## Step 1: the gradient is a Hermitian form

Write  $\psi = U_A \psi_0$ . Expanding  $\exp_p(p\theta A)$  to first order and using  $A^* = -A$ :

$$\nabla \mathcal{L} = p^{-1} \partial_\theta \mathcal{L} = \langle \psi, C \psi \rangle, \quad C = [\hat{M}, A] = \hat{M}A - A\hat{M}.$$

- $C$  is selfadjoint, so  $\nabla \mathcal{L} \in \mathbb{Q}_p$ ; all data integral gives  $\nabla \mathcal{L} \in \mathbb{Z}_p$ .
- The gradient is maximal  $\iff \nabla \mathcal{L}$  is a unit  $\iff$  its **leading digit**

$$\langle \tilde{\psi}, \Gamma \tilde{\psi} \rangle \in \mathbb{F}_p$$

is non-zero, where  $\sim$  denotes reduction modulo  $p$ .

The question is now about one Hermitian form over a *finite* field, evaluated at a random point.

## Step 2: Haar measure becomes counting

Compactness of  $G_N$  comes with a filtration by reduction:

$$G_N \xrightarrow{\pi_k} U_N(\mathcal{O}/\mathfrak{m}^k) \longrightarrow \cdots \longrightarrow U_N(\mathbb{F}_{p^2}).$$

### Pushforward

Each  $\pi_k$  is surjective (lift, then correct by Newton iteration), and sends  $\nu$  to the **uniform** measure on the finite group  $U_N(\mathcal{O}/\mathfrak{m}^k)$ .

### Transitivity

$G_N$  acts transitively on the integral sphere  $\{u : \langle u, u \rangle = 1\}$ , so  $\tilde{\psi}$  is **uniform** on the finite sphere  $\tilde{\mathbb{S}}_N = \{\tilde{u} \in \mathbb{F}_{p^2}^N : \langle \tilde{u}, \tilde{u} \rangle = 1\}$ .

This replaces the  $t$ -design combinatorics of the complex theory.

## Step 3: a character sum in closed form

In an orthonormal eigenbasis of  $\Gamma$ , with  $t_i = \text{Nm}(u_i) \in \mathbb{F}_p$ , the two constraints are  $\sum_i t_i = 1$  and  $\sum_i \lambda_i t_i = c$ . Count the solutions  $\mathcal{N}_c$  with an additive character  $\chi$  of  $\mathbb{F}_p$ . Everything reduces to the sum  $T(s) = \sum_{t \in \mathbb{F}_p} \#\text{Nm}^{-1}(t) \chi(st)$ , which is  $p^2$  for  $s = 0$  and  $-p$  for  $s \neq 0$ .

### Counting lemma

If  $\Gamma$  is  $r$ -non-degenerate, then for every  $c \in \mathbb{F}_p$

$$\left| \frac{\mathcal{N}_c}{|\widetilde{\mathcal{S}}_N|} - \frac{1}{p} \right| \leq 10 p^{1-r}, \quad |\widetilde{\mathcal{S}}_N| = p^{2N-1} - (-1)^N p^{N-1}.$$

Take  $c = 0$  (the leading digit vanishes) and the complement: that is the theorem.

## Side by side

|                | Complex                                | $p$ -adic                                         |
|----------------|----------------------------------------|---------------------------------------------------|
| Ensemble       | $U(2^n)$ , compact                     | full group non-compact; use $G_N$                 |
| Moments        | unitary $t$ -designs                   | character sums over $U_N(\mathbb{F}_{p^2})$       |
| Gradient       | $\text{Var} = O(2^{-n})$               | $\Pr[\text{maximal}] \geq 1 - p^{-1} - 10p^{1-r}$ |
| Scaling in $n$ | exponential decay                      | none                                              |
| Bad set        | full measure as $n \rightarrow \infty$ | locus $[\hat{M}, A] \equiv 0 \pmod{p}$            |
| Expressibility | more expressive, flatter               | more random, <b>larger</b> gradients              |

The last row is the one to remember: over  $K$ , randomness is what *produces* large gradients.

# What replaces the plateau

The obstruction does not vanish: it changes shape.

- **Complex:** a *measure* phenomenon. The bad set has full measure as  $n \rightarrow \infty$ ; no choice of  $M$  and  $A$  escapes it.
- **$p$ -adic:** a *locus*. The bad set is exactly  $[\hat{M}, A] \equiv 0 \pmod{p}$ : reduce two integer matrices and multiply them.
- As  $U_B$  varies, the locus is a clopen union of congruence classes of measure  $O(p^{-1})$ : small but **not null**, a distinction the complex theory never has to make.

## Design criterion

Choose  $\hat{M}$  and  $A$  that visibly fail to commute modulo  $p$ . The most ordinary pair (Pauli  $X$  against a phase rotation) already does.

# Does a maximal gradient imply trainability?

Not by itself.

- The theorem says the *leading digit* of the gradient is generically non-zero, so a step  $\theta \leftarrow \theta - \eta \nabla \mathcal{L}$  with  $|\eta|_p = 1$  moves the leading digit of  $\theta$ .
- Convergence is a separate condition. On a quadratic loss with Hessian  $H$ ,  $p$ -adic descent contracts only if  $\mathbf{1} - \eta H \equiv 0 \pmod{p}$ : with a scalar step,  $H$  must be **scalar modulo  $p$** .
- With a matrix preconditioner the condition relaxes to **invertibility of  $H$  modulo  $p$** , which is generic.
- When it holds, convergence is ultra-fast: error contracts by  $p^{-1}$  per step, so  $O(\log_p(1/\varepsilon))$  iterations.

We remove gradient concentration from the list of obstructions; trainability becomes a question of *conditioning*.

# Why study a model with no hardware?

No  $p$ -adic quantum computer exists. The result is still useful.

- ① **A diagnostic about the complex theory.** We keep the circuit, observable, ansatz and qubit count, change only the field, and the plateau disappears. The load is carried by compactness and by the Archimedean absolute value, which locates the hypothesis a mitigation has to attack.
- ② **A reusable method.** The pushforward reduces *any* Haar moment over  $G_N$  to counting in finite unitary groups: overlaps, expectation values,  $p$ -adic kernels.
- ③ **A checkable criterion.**  $[\hat{M}, A] \not\equiv 0 \bmod p$  is a finite test: no sampling, no asymptotics.

Applications would come through classical  $p$ -adic learning on hierarchical data, where the geometry is ultrametric to begin with.

# Limitations

- **The splitting hypothesis is real.** Over a finite field a Hermitian matrix may have eigenvalues outside  $\mathbb{F}_p$ ; we give a  $2 \times 2$  counterexample over  $\mathbb{F}_9$ . Outside the splitting class the question is open.
- **Level one only.** We control  $\Pr[v_p(\nabla \mathcal{L}) \geq 1]$ , not the full valuation distribution (expected:  $\Pr[v_p \geq k] = p^{-k}(1 + o(1))$ , not proved).
- **Standing assumptions.**  $p$  odd,  $K/\mathbb{Q}_p$  unramified, one parameter in a two-block ansatz.
- **No experiments.** No sampling-based readout exists. The useful companion is exact arithmetic in  $\mathcal{O}/\mathfrak{m}^k$  for small  $n$  and  $p$ , checking the counting lemma by enumeration.

# Closing

Barren plateaus are a theorem about the  
Archimedean absolute value.

Change the field and they go.

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## Backup: why the splitting hypothesis is not free

Over  $\mathbb{F}_9 = \mathbb{F}_3(\iota)$ ,  $\iota^2 = -1$ , take

$$\Gamma = \begin{pmatrix} 0 & 1 + \iota \\ 1 - \iota & 0 \end{pmatrix}, \quad \chi_\Gamma(x) = x^2 - \text{Nm}(1 + \iota) = x^2 + 1.$$

- $\Gamma$  is Hermitian, but its eigenvalues  $\pm\iota$  lie in  $\mathbb{F}_9 \setminus \mathbb{F}_3$ .
- If  $\Gamma v = \lambda v$  with  $\lambda \neq \bar{\lambda}$ , then  $\lambda \langle v, v \rangle = \bar{\lambda} \langle v, v \rangle$  forces  $\langle v, v \rangle = 0$ : the eigenvector  $(1 + \iota, \iota)$  has norm  $2 + 1 = 0$ .
- No orthonormal eigenbasis exists.

Closing the gap needs the classification of *pencils* of Hermitian forms over a finite field: the main open follow-up.

## Backup: how large can $r$ be?

$\Gamma$  has  $N = 2^n$  eigenvalues in the  $p$ -element field  $\mathbb{F}_p$ , so pigeonhole gives  $m_{\max} \geq \lceil N/p \rceil$  and

$$r \leq N - \lceil N/p \rceil \leq N \left(1 - \frac{1}{p}\right).$$

- A simple spectrum is impossible once  $N > p$ .
- Spread evenly,  $m_{\max} \approx N/p$  and  $r \approx N(1 - 1/p)$ .
- The witness spreads them over two values:  $r = N/2$ .

Either way  $r = \Theta(2^n)$ , so  $10p^{1-r}$  is of order  $p^{-2^n}$ .

## Backup: the bad set is a commutator

### Corollary

If  $\Gamma$  splits, it fails to be 1-non-degenerate if and only if  $\hat{M}$  and  $A$  commute modulo  $p$ .

- Failure means  $m_{\max} = N$ , i.e.  $\Gamma = \lambda \mathbf{1}$ .
- A commutator has trace 0, so  $N\lambda = 0$  in  $\mathbb{F}_p$ .
- $N = 2^n$  and  $p$  is odd, so  $p \nmid N$  and  $\lambda = 0$ .
- Hence  $\Gamma = 0$ , i.e.  $[\hat{M}, A] \equiv 0 \pmod{p}$ .

## Backup: surjectivity of reduction (Newton step)

Lift  $g \in U_N(\mathcal{O}/\mathfrak{m}^k)$  to an integral  $g_0$  with  $g_0^* g_0 = \mathbf{1} + p^l S$ ,  $S^* = S$ . Put  $T = -S/2$  (integral,  $p$  odd) and  $g_1 = g_0(\mathbf{1} + p^l T)$ . Then

$$g_1^* g_1 = \mathbf{1} + p^l (T + S + T) + O(p^{2l}) = \mathbf{1} + O(p^{2l}).$$

- Each step doubles the precision and does not change  $g_0$  modulo  $\mathfrak{m}^k$ .
- The limit  $h$  is integral with  $h^* h = \mathbf{1}$ , so  $h \in G_N$  and  $\pi_k(h) = g$ .
- A continuous surjective homomorphism of a compact group carries Haar measure to the uniform measure on the finite image.